

Q-1:

$$\begin{aligned}\text{Cache index} &= 129 \bmod 4 \\ &= 1000\ 0001 \bmod 2^2 \\ &= 01 \text{ (Ans)}\end{aligned}$$

Q2

$$\text{block address} = \left\lfloor \frac{321}{10} \right\rfloor = \lfloor 32.1 \rfloor = 32 = 10\ 0000 \text{ (Ans)}$$

Q3

$$\text{Block Size} = 32 \text{ bits} = 4 \text{ bytes}$$

$$\text{Cache Size} = 64 \text{ bits} = 8 \text{ bytes} = \frac{8}{4} = 2 \text{ blocks}$$

$$\text{Block Add.} = \left\lfloor \frac{239}{4} \right\rfloor = \lfloor 59.75 \rfloor = 59$$

$$\begin{aligned}\text{Index} &= 59 \bmod 2 \\ &= 1 \text{ (Ans)}\end{aligned}$$

Q-4

$$\begin{aligned}\text{Block Size} &= 8 \text{ bits} = 1 \text{ byte} \\ \text{Cache Size} &= 32 \text{ bits} = 4 \text{ bytes} = \frac{4}{1} = 4 \text{ blocks} \\ \text{Index bit size} &= \log_2(4) = 2 \text{ bits} \\ \text{Offset} &= \log_2(1) = 0 \text{ bits} \\ \text{Tag} &= (9 - 2 - 0) = 7 \text{ bits}\end{aligned}$$

Address

$$= 297$$

$$= 100101001$$

0 bits

$$\text{Address} = \underbrace{100101001}_{\text{Tag bits}} \underbrace{}_{\text{Index bits}}$$

$$\text{Index bits} = 01$$

$$\text{Tag bits} = 1001010$$

$$\text{No offset bits (Ans)}$$

Q-5

Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
010	1	011	Data
011	1	001	Data
100	0		
101	1	000	Data
110	1	010	Data
111	0		

$$\begin{aligned}\text{From the table, block size} &= 8 \text{ bits} = 1 \text{ byte} \\ \text{Cache} &= 8 \text{ blocks} \\ \therefore \text{Index size} &= \log_2(8) = 3 \text{ bits} \\ \therefore \text{Offset} &= \log_2(1) = 0 \\ \therefore \text{Tag} &= (6 - 3) = 3 \text{ bits}\end{aligned}$$

a)

	Mem. Add.	hit/miss
1	<u>100</u> <u>100</u>	miss
2	<u>111</u> <u>010</u>	miss
3	<u>010</u> <u>110</u>	hit
4	<u>100</u> <u>100</u>	hit
5	<u>111</u> <u>110</u>	miss

$$\text{b) Hit Ratio} = \frac{2}{5}$$

$$\text{Miss} = 1 = \frac{3}{5}$$

$$\begin{aligned}\text{c) AMAT} &= \text{Hit time} + \text{Miss Rate} \times \text{Miss Penalty} \\ &= 2 + \frac{3}{5} \times 5 = 5\end{aligned}$$

$$\text{d) } 20 \rightarrow 1 \text{ cycle}$$

$$10 \rightarrow \frac{1}{2} \text{ cycle}$$

$$\therefore 50 \rightarrow \frac{5}{2} \text{ cycle} = 2.5 \text{ cycle}$$

So, 2.5 cycles are required per instruction

Updated cache table after access 1 -

Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
✓ 010	1	011	Data
011	1	001	Data
100	0 1	100	Data
101	1	000	Data
110	1	010	Data
111	0		

Updated cache table after access 2 -

Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
✓ 010	1 1	0 111	Data Data
011	1	001	Data
✓ 100	0 1	100	Data
101	1	000	Data
✓ 110	1	010	Data
111	0		

As, access-3 was a hit, so cache table remains the same.

Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
✓ 010	1 1	0 111	Data Data
011	1	001	Data
100	0 1	100	Data
101	1	000	Data
✓ 110	1	010	Data
111	0		

As, access-4 was a hit, so cache table remains the same.

Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
✓ 010	1 1	0 111	Data Data
011	1	001	Data
100	0 1	100	Data
101	1	000	Data
✓ 110	1	010	Data
111	0		

Updated cache table after access 5 -

Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
✓ 010	1 1	0 111	Data Data
011	1	001	Data
100	0 1	100	Data
101	1	000	Data
✓ 110	1 1	0 111	Data Data
111	0		